Lab 05

Due March 3 5:00 PM PST

STAT240 Spring 2026

2026-02-27

For this homework, provide a single rendered Quarto Markdown file in pdf format on crowdmark for the problems (you may render the qmd file to html, and then convert the html file to pdf using the print function on your web browser). You can use the template `Lab_05.qmd`. **Provide in the markdown the final version of all of the code you wrote for this homework, and make sure long lines of code are wrapped in the rendered pdf. Lines which are cut off cannot be graded. If your Markdown file involves examining a large dataset, do not print the entire dataset to the markdown file in the steps of your solution (instead, suppress the output, or only show a small section of the data as an example).**

Problem 1

This problem is about bird sightings in Canada. The files `ebird30.csv` and `names.csv` contain data contributed by citizen scientists in 30 days to eBird: A resource maintained by the Cornell Lab of Ornithology. The file ebird30.csv is a comma separated variable file with headers, and one row per bird species. The first column is named Code and contains the code for each bird species available in the data (these codes were created by eBird). The remaining columns are named according to the provinces and territories (regions) of Canada (for example, CA.NU is Nunavut). For each bird species and region, the corresponding entry of ebird30.csv indicates the number of birds of that species observed in that region in the past 30 days. The file names.csv is a comma separated variable file with headers, and one row per bird species. The first column is the code for a species of bird, and the remaining columns are the common name and scientific name for that bird. Included in the archive for this lab is an R script `canada.R`. This script contains a function `canada` that will make a heatmap of Canada, showing values for each province. This script requires that several packages be installed (they can be installed using the script `packages.R`)

Problem 1a)

How many bird species have been spotted in only one region of Canada? Verify that your number is correct by using the Canada function to plot the counts of one the bird species you identify as being spotted in only one region of Canada (all regions should be dark blue except one). Provide the plot.

(4 points)

Problem 1b)

Which bird species has the highest count, summed across all regions? Give the common name, and the scientific name.

(2 points)

Problem 1c)

Which bird species has the smallest non-zero count, summed across all regions? If there are more than one, provide the number of bird species that are tied for smallest non-zero count, and sort their common names in alphabetical order and list the first 10. (Hint: The sort function applied to a vector of strings will return a sorted version of the strings in alphabetical order.)

(4 points)